MILITARY SPECIFICATION

SHEILD, RAD HAz

L/20MM CARTRIDGES

HOLDER, RAD HAZ SHIELD

This Specification has been approved
by the Bureau of Naval Weapons,
Department of the Navy.

1. SCOPE

1.1 This specification covers the rad haz shield, hereinafter referred to as “shield” and the rad haz shield holder, hereinafter referred to as “holder”. The shield series as a protective cover for the electric primer of a 20mm MARK 100 series round of ammunition belted in the MARK 2 link.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

W-L-800	Lubricating Oil, General Purpose, Preservative (Water-Displacing, Low Temperature).

Military

MIL-L-16727	Link, Ammunition, 20mm, MARK 2.
MIL-P-16232	phosphate Coatings, Heavy, Manganese or Zinc Base (For Ferrous Metals)

Source: https://assist.dla.mil -- Downloaded: 20261O-23T05:07Z
Check the source to verify that this is the current version before use.

sTANDARDS

Federal

Federal Test Method Standard No. 151	Metals; Test Methods.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

PUBLICATIONS

Bureau of Naval Weapons OP 1866	20mm Aircraft Gun, MARK 12 Mods 0, 2, and 3, Description, Operation and Maintenance.

DRAWINGS

Bureau of Naval Weapons (Code Ident 10001)

2164723	Shield, Rad Haz.
2164725	Holder; Rad Haz Shield.
2164728	Shield Assembly.
LD 534529	Shield Packaging (All Drawings, LD's, Specifications, and Standards Listed thereon).
LD 534561	Primer Shield Installation Device.

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring agency, or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preporduction lot. Unless otherwise specified i the contract or order, a preproduction lot of shields and holders shall be manufactured using the methods and procedures proposed for the production. The pre-production lot shall be tested as specified in Section 4 herein and is for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract (See 6.2).

3.1.1 The procuring activity may waive the preproduction lot requirement of 3.1 provided that the supplier has demonstrated his ability to conform with the requirements of this specification under previous

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:07Z
Check the source to verify that this is the current version before use.

contracts covering manufacture of shields and holders.

3.2 Compliance with documents. Unless otherwise specified, the shields and holders shall be in accordance with the requirements specified herein and in the applicable documents listed in Section 2.

3.3 Materials. Materials used in the manufacture of shields and holders shall to the material specifications listed in the applicable drawings (See Section 2).

3.4 Design. Shields and holders shall conform with the design requirements specified by the applicable drawings and this specification (See Section 2).

3.5 Construction. The shields and holders shall meet the require ments specified in 3.5.1, 3.5.2, and 3.5.3.

3.5.1 Link compatibility. The shields shall be compatible with the 20mm MARK2, MODS 0, 1, and 2, ammunition links, specification MIL-L-16727. Subsequent to installation onto the links, shields shall not bend, crack, split, or detach from the links during handling, loading, firing, or ejection (See 4.6.2).

3.5.2 Primer protection. Subsequent to shield instalation onto belted MARK 100 series electric primed ammunition, the shield protective tab shall completely cover the primer during handling and loading (See 4.6 .2).

3.5.3 Primer shield installation device compatibility. Shield-holder assemblies shall be compatible with the primer shield installation device, LD 534561 (See 4.6.2.4).

3.6 Finish.

3.6.1 Burrs. All burrs should be removed from shields and holders. Cutting, stamping, and forming operations shall be performed in such a manner that all burrs which may be formed will be directed toward the outside of the shield. No burrs shall be permissible on the protective tab of the shield or retaining tabs of the holder.

3.6.2 Protective coating. Subsequent to appropriate hardness treat- ment, both shields and holders shall receive the protective coating specified by the applicable drawings (See Section 2).

3.7 Environmental requirements.

3.7.1 Resistance to salt spray. Each shield and holder tested in accordance with 4.6.1.2 shall be examined visually for evidence of rust. Each shield shall show no more than four easily discernible rust spots, each measuring not more than 1/8 inch in diameter or equivalent area. Each holder shall show not more than eight easily discernible rust sp;ots of the specified area. Rust staining, rust drainage, and edge rust shall not be considered when measuring the area of rust spots.

Source: https://assist.dla.mil -- Downloaded: 202610-23T05:07Z
Check the source to verify that this is the current version before use.

3.7.2 Temperature extremes. The shields shall comply with the requirements of 3.5.1 and 3.5.2 at any temperature between minus 70 degrees Fahrenheit (F) and plus 165 degrees F.

3.8 Government-furnished property. Ammunition boxes, LD 165132, in quantities specified In the contract or order, are required to comply with the packaging requirements of Section 5.

3.9 Government-loaned property. The following are required to determine shieLd conformance with the requirements of this specification:

(a) Dummy Rounds, 20mm, MARK 103, MOD 0.
(b) Links, 20mm, MARK 2, MODS 0, 1, and 2.

3.10 Packaging, packing, and marking. Packaging, packing, and marking of the shield assemblies shall conform to the requirements of Section 5 of this specification.

3.11 Workmanship. Shields and holders shall be uniform in quality and temper, and shall be free from laminated metal (complete separation of the metal at some poimt)s excessive scale, scraps, splits, cracks, seam, and other defects. The quality of workmanship shall assure conformance to all requirements of this specification and the applicable docuuqmts listed in Section 2. The standards of workmamhip exhibited in any approved pre-production sample, subject to any qualification stated in the Government notice of approval, shall be determinative of the requirements of the contract relative to workmanship in so far as not specifLcally covered by applicable specifications.

3.12 Data. No data is required by this specification, or by applicable documents referenced in Section 2, unless specified in con-tract or order. (See paragraph 6.2.)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the per-formance of all inspections specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perorm any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection and testing of the shields and holders shall be classified as follows:

(a) Preproduction inspection.
(b) Quality conformance inspection.

4.3 Lot. A lot shall consist of all shields and holders manufactured by the same press at the same plant locationa, and submitted for Government

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:07Z
Check the source to verify that this is the current version before use.

acceptance at the same time, except that the lot size shall not exceed 20,000 shields and 2000 holders.

4.4 Sampling.

4.4.1 Preproduction sample. A preproductlon sample of 600 shields and 70 holders manufactured in accordance with 3.1 shall be subjected to the following tests and inspections:

Examination or Test	Requirement Paragraph	Method Paragraph	No. of Sample Units to be Inspected	No. of Defective Allowed
Group I				
Material, Design, and Workmanship	3.2, 3.3, 3.4, 3.6, and 3.11	4.6.1.1	40 shields 10 holders	0 0
Group II				
Salt Spray	3.7.1	4.6.1.2	10 shields 5 holders	0 0
Group III				
Bench Tests	3.5.1 and 3.5.2	4.6.2.5	150 shields	See 4.6.2.7.1
Group IV				
Flight Tests	3.5.1 and 3.5.2	4.6.2.6	400 shields	See 4.6.2.7.2

Acceptance of the preproduction sample shall be based on no defects. Any production by the supplier prior to approval of the preproduction lot shall be at the supplier's risk.

4.4.2 Quality conformance inspection sampling. The sampling plans and procedures set forth in Standard MIL-STD-105 shall be used in the determination of acceptability of the shields and holders tested and . inspected as specified in 4.5 and 4.6.1.

4.5 Examination.

4.5.1 Components. Prior to assembly of the shield and holder, it shall be ascertained that all components procured under separate documents have been inspected, tested, and accepted in accordance with their respective documents.

4.5.2 Packing and marking. It shall be ascertained that packing and marking of the shield assemblies conform to the requirements of Section 5 of this specification.

4.6 Tests 1

Source: https://assist.dla.mil -- Downloaded: 2026l0-23T05:07Z
Check the source to verify that this is the currentversion before use.

4.6.1 Acceptance tests.

4.6.1.1 Visual and mechanical inspection. Shield and holder samples selected in accordance with 4.4.2 shall be tested for confozmsnce to 3.2, 3.3, 3.6, 3.10, and 3.11. Lot quality shall be judged in accordance with Standard MIL-STD-105, Inspectim Level II, Acceptable Quality Level (AQL) 1.0 percent defective for major defects, plus 2.5 percent defective for minor defects.

4.6.1.2 Salt spray resistance test. After the shields and.holders have received the phosphate coating of Specification MIL-P-162, samples shall be selected in accordance with MIL-STD-105, Inspection level S-4, acceptance number zero and subjected to the best of Federal Test Method Standard No. 151, Method 811.1, for 48 hours. The shields and holders shall meet the requirements of 3.7.1. Test shields and holders shall not be reused for other tests, nor shall they be delivered to the Government as units of production.

4.6.2 Functioning tests.

4.6.2.1 Reproduction sample. From each preproduction lot$_a$ 55 shield assemblies, (550 shields and 55 holders), drawing 2164728, shall be delivered to a Naval-test activity (as designated by the Bureau of Naval Weapons), for functioning tests (See 6.2).

4.6.2.2 Reduction sample. From each production lot 15 shield assemblies (150 shields and 15 holders), drawing 2164728, shall be delivered to a Naval test activity (as designated by the Bureau of Naval Weapons), for functioning tests (See 6.2).

4.6.2.3 The functioning tests shall include flight tests (firing shield equipped ammunition from air-borne Navy aircraft), and bench tests (firing shield equipped ammunition from bench test mount).

4.6.2.4 The primer shield installation device shall be used for installing shields onto the links prior to all functioning tests.

4.6.2.5 Bench test. The bench test shall include the following:

(a) One group of shields to be cold conditioned at minus 70 degrees F for at least two hours immediately prior to test firing; a second group of shields from the same lot to be hot conditioned at plus 165 degrees F for at least two hours immediately prior to test firing, and a third group of shields from the same lot to be conditioned at plus 70 degrees F. Shields shall be conditioned subsequent to installation onto belted ammunition.

(b) Use of new MARK 2 MODS 0, 1, and 2 links, and MARK 105 TP rounds of ammunition.

(c) Use of MARK 12, MOD 0, gun equipped with the MARK 7 MOD 2 feed mechanism.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:07Z
Check the source to verify that this is the current version before use.

(d) Feeding ammunition in both right and left hand directions.
(e) Ammunition and shields to be lightly oiled with lubricating oil confordng with VV-L-800.
(f) The flexible feed chute shall-be unsupported for the 18 inches nearest the feed mechanism. The supported part of the flexible feed chute shall be on the same level as the entrance to the feed mechanism. There shall be enough slack in the feed chute so that it may be raised at the middle of the unsupported length to a height of 8 inches above the lowest point (determined under slight load). This shall give a dynamic belt whip of at least 4 inches total amplitude.
(g) Rate of fire to be at least 925 rounds per minute.
(h) Use of link ejection chutes comparable with those in current Navy aircraft having MARK 12 gun installations.

4.6.2.5.1 Test usingMARK2, MOD O links, plus MARK 2, MOD 1, end links.

4.6.2.5.1.1 Right hand feeding. Test fire 75 shield equipped rounds under the following conditions:
(a) Fire one group of 25 rounds equipped with cold conditioned shields, (-70°F).
(b) Fire a second group of 25 rounds equipped with shields conditioned at ambient temperature, (+70°F).
(c) Fire a third group of 25 rounds equipped with hot conditioned shields, (+165°F).

4.6.2.5.1.2 Left hand feeding. Repeat the tests prescribed in 4.6.2.5.1.1, reusing the same shields, but firing from left hand feed.

4.6.2.5.1.3 Shields conditioned at one temperature shall be re-conditioned at that same temperature> and shall not be intermixed with other groups.

4.6.2.5.1.4 Each shield shall be used a total of two times for the tests of 4.6.2.5.1.1 and 4.6.2.5.3..2.

4.6.2.5.2 Test using MARK2, MOD2 links, plus MARK2, MOD 1, end links.

4.6.2.5.2.1 Right hand feeding. Repeat the tests prescribed in 4.6.2.5.1.1, but using 75 new shields not previously tested, and MARK 2, MOD 2 links, plus MARK2 MOD 1 end links.

4.6.2.5.2.2 Left hand feeding. Repeat the tests prescribed in 4.6.2.5.1.2, but re-using the same shields used in the test of 4.6.2.5.2.1.

4.6.2.5.4 Each shield shall be used a total of two times for the tests of 4.6;2;5.2.1 and 4.6.2.5.2.2.

4.6.2.5.4 When shield failures occur which cause a lot to be rejected, testing of that lot shall be terminated. If shield breakage or link-shield separations (partial or total detachment of the shield from the link) occur which do not cause rejection of the lot, remove the broken or dislodged shields and continue testing (See 4.6.2;7). Records of shield breakages and link-shield separations shall be kept for each lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:07Z
Check the source to verify that this is the current version before use.

4.6.2.6 Flight tests.

4.6.2.6.1 Preproduction lot. Four hundred shields from each preproduction lot (4.6.2.1) shall be subjected to flight tests (See 4.6.2.6.3).

4.6.2.6.2 Production lot. When requested by the procuring activity, the contractor shall furnish 400 shields from each production lot (not a part of the samples submitted in accordance with 4 .6.2.2), for shipment to a Naval test activity (as designated by the Bureau of Naval Weapons), for flight tests. These shields shall be furnished at no cost to the Government. No production lot shields shall be submitted for flight tests unless specified in the contract, order, or by the procuring activity (See 6.2).

4.6.2.6.3 Flight test waiver. The procuring actitity may waive the flight tests requrements of 4.6.2.6.1 provided that the contractor has demonstrated his ability to conform with the requirements of this specification under previous contracts covering manufacture of shields. No preproduction lot shields for flight tests shall be delivered to the testing activity when the procuring activity waives the requirements of 4.6.2.6.1 (See 6.2).

4.6.2.6.4 Test requisitions. The flight tests shall include the following:

(a) Use of a current operational Navy aircraft equipped with at least one MARK 12, 20mm gun known to have a cyclic rate of at least 925 rounds per minute.
(b) Use of MARK 2, MOD 0 and/or MARK 2, MOD 2 links, plus MARK 2, MOD 1 end links.
(c) Ammunition to be lightly oiled with lubricating oil conforming with W-L-800 or equivalent.
(d) Firing from one or more MARK 12 MOD 0 guns equipped with the MARK 7 MOD 2 feed mechanism.

4.6.2.6.5 At least two hundred rounds of shield equipped ammunition shall be burst fired while the aircraft is subjected to three to five “g's” and an air speed of at least 500 knots.

4.6.2.7 Acceptance criteria.

4.6.2.7.1 Lot disposition for the tests of 4.6.2.5 shall. be as follows:
Step 1. The entire lot shall be rejected if one shield breaks, or one link-shield separation occurs during the first use of the shields. If no shield breakage or link-shield separation occurs, then proceed to Step 2.
Step 2. The entire lot shall. be rejected if two or more shields break, or two or more link-shield separations occur during the second use of the shields. One shield breakage and one link-shild separation in combination will also reject the lot.

4.6.2.7.2 Lot disposition for the tests of 4.6.2.6 shall be as follows:
The entire lot shall be rejected for failure-to-fire resulting from one or more shield breakages or one or more link-shield separations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:07Z
Check the source to verify that this is the current version before use.

4.6.2.7.3 Link-shield separations and shield breakages caused by faulty links (broken, cracked, deformed, etc.), faulty link chutes, and other defective testing equipment shall. not be counted as failures of the shields during the tests prescribed in 4.6.2.

4.6.2.7.4 Link-shield separations and shield breakages occurring subsequent to ejection from the link chute shall not be used as criterion for failure of the shields during the tests prescribed in4.6.2.5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, or C as specified (See 6.2).

5.1.1 Level A.

5.1.1.1 Preservation. The supplemental oil finish applied subsequent to phosphate coating, as specified by the applicable drawings, shall serve as the preservative for the shields and holders. No additional preservative is required.

5.1.1.2 Unit Packaging. Not applicable.

5.1.2 Level C.

5.1.2.1 Preservation. Preservation shall be as specified in 5.1.1.1.

5.1.2.2 Unit packaging. Not applicable.

5.2.1 Level A.

5.2.1.1 Exterior containers. Shield assemblies, in numbers specified in the contract or order (See 6.2), shall be packed into containers conforming to drawings listed in ID 534529.

5.2.1.2 Cushioning. The pack shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices, tO ensure that the shield will meet all of the requirements of section 3.

5.2.2 Level C.

5.2.2.1 Exterior containers. Shield assemblies, in numbers specified in the contract or order (see 6.2), shall be packed to afford p rotection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall conform to the carrier rules and regulations applicable to the mode of trasmportion.

5.2.2.2 Cushioning. Cushioning shall be as specified in 5.2.1.2.

Source: https://assist.dla.mil -- Downloaded: 2026t0-23T05:07Z
Check the source to verify that this is the currentversion before use.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intednded use. The shield is designed for clip-on installation onto 20mm, MARK 100 series electric primed ammunition, belted in MARK 2 links. Ten shields are assembled in each holder and are simultaneously installed, directly from the holder by the hand operated primer shield installation device, onto the after ends of the link-round asseniblies in the belt of ammunition. Holders are discarded after the shields are installed. The shield serves as a protective cmer for the exposed electric primer during handling, magazine loading, etc. or as the round travels through its chuting to the gun. By completely covering the primer, the shield prevents it from contacting any surface which is irradiated by radio-frequency electromagnetic fields, and thereby prevents accidental ignition. The rounds of ammunition are stripped from their links by the feed mechanism and chambered while the link-shield assemblies continue through the feed mechanism and link ejection chute where they are expended.

6.2 Ordering data.Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Size of preproduction lot if different from 4.4.1.
(c) Facility designated to evaluate the preproduction lot.
(d) Flight testrequirement of the preproduction lot (See 4.6.2.6.3).
(e) Goverment-furnished property (See 3.8).
(f) Government-loaned property (See 3 .9).
(g) Lot size if different from 4.3.
(h) Facility designated to evaluate the production lots.
(i) Flight test requirements of the production lot (See 4.6.2.6.2).
(j) Procedure for resubmittal of rejected lots.
(k) Preservation, packaging and packing levels.
(l) Special marking of shipping containers if required.
(m) Data (see 3.12).

Source: https://assist.dla.mil -- Downloaded: 2026t0-23T05:07Z
Check the source to verify that this is the currentversion before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-S-81245(WP)SHIELD, RAD HAZ L/20MM Cartridges Holder, Rad Haz Shield

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 83 1426

C-8279

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:07Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
Bureau of Naval Weapons
Washington, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

BUREAU OF NAVAL WEAPONS (RREN-5)
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:07Z
Check the source to verify that this is the current version before use.

INCH-POUND

NOTICE OF INACTIVATION FOR NEW DESIGN

MIL-S-81245(WP)
NOTICE 1
21 August 1989

MILITARY SPECIFICATION

SHIELD RAD HAZ L/20MM CARTRIDGES HOLDER,
RAD HAZ SHIELD

This notice should be filed in front of MIL-P-81245(WP)
dated 21 April 1965

MIL-S-81245(WP) is inactive for new design and is no longer used by the Navy except for replacement purposes. It may also be required for foreign military sales spares procurement.

Preparing activity:
Navy - AS
(Project 1005-N731)

AMSC N/A

FSC 1005

DISTRIBUTION STATEMENT A. Approved for public release; distribution Is unlimlted.

Source: https://assist.dla.mil -- Downloaded: 2026l0-23T05:07Z
Check the source to verify that this is the current version before use.